KENYA INSTITUTE OF CURRICULUM DEVELOPMENT

A Skilled and Ethical Society

DRAFT

SENIOR SCHOOL CURRICULUM DESIGN

GRADE 10

POWER MECHANICS

2025

First Published in 2025

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FOREWORD

The Government of Kenya (GoK) is committed to ensuring that policy objectives for Education, Training and Research meet the aspirations of the Kenya Constitution 2010, the Kenya Vision 2030, National Curriculum Policy 2019, the United Nations Sustainable Development Goals (SDGs) and the Regional and Global conventions to which Kenya is a signatory.

Towards achieving the mission of Basic Education, the Ministry of Education (MoE) embarked on a reform process that culminated in the adoption of a Competency Based Education (CBE) system in 2017. This came after extensive research, stakeholder engagement international benchmarking and consideration of the 21st Century socio-economic realities. CBE in Kenya adopted a 2-6-3-3 structure where the numbers indicate the years learners take at various levels in Basic Education (Pre-Primary, Primary, Junior School and Senior School; respectively). CBE has for critical elements: The Competency Based Curriculum (CBC), Pedagogy (teaching and learning), Resources and the Competency Based Assessment. The MoE oversees the provision of resources while the other components (Curriculum, Pedagogy and Assessment are handled by the Kenya Institute of Curriculum Development, (KICD), Teachers Service Commission, (TSC) and the Kenya National Examinations Council (KNEC); respectively.

KICD has successfully developed curriculum designs that have been used in the implementation of the Competency Based Curriculum (CBC) for the Pre-Primary, Primary and Juniors School levels. The first cohort of learners pursuing the CBC will transit to Senior School (Grades 10-12) in 2026. The curriculum designs at this level are expected to facilitate learners to acquire relevant knowledge, skills, values and attitudes (competencies) in preparation for tertiary education and the world of work. I urge all agencies of the GoK and other stakeholders in Education to use the designs to plan for effective and efficient implementation of Competency Based Education.

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PREFACE

The Ministry of Education (MoE) through the State Department for Basic Education, has been implementing the Competency Based Education (CBE) since the adoption of the *Basic Education Curriculum Framework* (KICD, 2017) during the national convention in January 2017. The Kenya Institute of Curriculum Development (KICD) developed Competency Based Curriculum (CBC) designs that were used for pilot studies in 2017 and 2018 before the roll-out at Pre-Primary and Primary Early Year (Grades 1 to 3 in 2019. The curriculum was then introduced in the subsequent grades through *a phase in –phase out* strategy, to ensure a smooth transition. Currently, the first cohort of CBC learners are at Grade 9 – the last grade at Junior School. The implementation of CBE has since benefited immensely from the recommendations of the *Presidential Working Party on Education Reforms* (PWPER) that were released in August, 2023 and fully implemented, in terms of curriculum designs.

Grade 10 is the entry class of the Senior School (SS) in Basic Education, a very critical level where learners shall have the opportunity to focus on pathways, tracks and subjects of their choices guided by the potentials demonstrated at earlier levels. The Senior School shall; therefore, form a foundation for further education and training and/or initial competencies for the Industry. This is an important step in the realisation of the Vision and Mission of the curriculum reforms as enshrined in the Sessional Paper No. I of 2019 whose title is: *Towards Realizing Quality, Relevant and Inclusive Education and Training for Sustainable Development* in Kenya.

The Grade 10 curriculum designs, like the designs for earlier levels, are anchored on the Kenyan National Goals of Education. They also present essence statements, general and specific expected learning outcomes as well as strands and sub strands for the learning areas. The designs also outline suggested learning experiences, key inquiry questions, core competencies, Pertinent and Contemporary Issues (PCIs), values and suggested assessment.

The Ministry of Education, through the State Department for Basic Education commits to provide resources and relevant guidelines for a smooth implementation of the CBC. It is expected that the designs will guide teachers to effectively facilitate learners to attain the expected learning outcomes for Grade 10 specifically and Senior School in general.

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ACKNOWLEDGEMENT

The Kenya Institute of Curriculum Development (KICD) Act Number 4 of 2013 (Revised 2019) mandates the Institute to develop curricula, programmes and curriculum support materials for Basic and Tertiary Education and Training. The curriculum development process involves research, international benchmarking and robust stakeholder engagement. Through a systematic and consultative process, the KICD conceptualised the Competency Based Curriculum (CBC) as captured in the *Basic Education Curriculum Framework* (KICD,2017), that responds to the demands of the 21st Century, the United Nations Sustainable Development Goals (SDGs). African Agenda 2063, East African Community Protocol and the aspirations captured in the Kenya Constitution 2010 and the Kenya Vision 2030.

KICD receives its funding from the GoK to enable the successful achievement of the stipulated mandate and implementation of the Government and Education sector plans. The Institute also receives support from development partners targeting specific programmes. The Grade 10 curriculum designs were developed with the support of the World Bank and GoK through the *Kenya Primary Education Equity in Learning Programme* (KPEELP); a project coordinated by the MoE. Therefore, the Institute is very grateful to the GoK, through the MoE and the development partners (especially The World Bank) for support in terms of policy guidance, resource provision and capacity building. Specifically, I wish to express special thanks to the Cabinet Secretary-MoE and the Principal Secretary-State Department for Basic Education.

We also wish to acknowledge the KICD curriculum development officers and other staff, all teachers, educators who took part as panellists; the institutions that participated and representatives of various stakeholders for their roles in the development of the Grade 10 curriculum designs. In relation to this, we acknowledge the support of the Chief Executive Officers of the Teachers Service Commission (TSC), the Kenya National Examinations Council (KNEC) and other agencies in the sector for their support in the process of developing these designs.

Finally, we are very grateful to the KICD Council Chairperson and other members of the Council for consistent guidance and oversight. We assure all teachers, parents and other stakeholders that these curriculum designs will effectively guide the implementation of the CBE at Grade 10 and preparation of learners for transition to further education and training or the world of work.

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NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. **Foster nationalism and patriotism and promote national unity.**
   Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.

2. **Promote the social, economic, technological and industrial needs for national development.**
   Education should prepare the youth of the country to play an effective and productive role in the life of the nation.
   a) **Social Needs**
      Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution following in the wake of rapid modernization. Education should assist our youth to adapt to this change.
   b) **Economic Needs**
      Education in Kenya should produce citizens with the skills, knowledge, expertise and personal qualities that are required to support a growing economy. Kenya is building up a modern and independent economy which is in need of an adequate and relevant domestic workforce.
   c) **Technological and Industrial Needs**
      Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognizes the rapid industrial and technological changes taking place, especially in the developed world. We can only be part of this development if our education system is deliberately focused on the knowledge, skills and attitudes that will prepare our young people for these changing global trends.

3. **Promote individual development and self-fulfilment**
   Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential interests and abilities. A vital aspect of individual development is the building of character.

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4. **Promote sound moral and religious values.**
   Education should provide for the development of knowledge, skills and attitudes that will enhance the acquisition of sound moral values and help children to grow up into self-disciplined, self-reliant and integrated citizens.

5. **Promote social equity and responsibility.**
   Education should promote social equality and foster a sense of social responsibility within an education system which provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability or geographical environment.

6. **Promote respect for and development of Kenya's rich and varied cultures.**
   Education should instil in the youth of Kenya an understanding of past and present cultures and their valid place in contemporary society. Children should be able to blend the best of traditional values with the changing requirements that must follow rapid development in order to build a stable and modern society.

7. **Promote international consciousness and foster positive attitudes towards other nations.**
   Kenya is part of the international community. It is part of the complicated and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities, rights and benefits that this membership entails.

8. **Promote positive attitudes towards good health and environmental protection.**
   Education should inculcate in young people the value of good health in order for them to avoid indulging in activities that will lead to physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

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THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM

Senior School is the fourth level of Basic Education in the Competency Based Curriculum (CBC) that learners shall proceed to after the Pre-Primary School, Primary School and Junior School levels. The essence of Senior School is to offer learners a pre-tertiary/university/ pre-career experience in which the learners have an opportunity to choose pathways, tracks and subjects where they shall have demonstrated interest and/or potential at the earlier levels. Senior School comprises three years of education for learners generally in the 16-18 years age bracket and lays the foundation for further education and training at the tertiary level and the world of work. In the CBC vision, learners exiting this level are expected to be ***engaged, empowered and ethical citizens*** ready to participate in the socio-economic development of the nation.

At this level, learners shall take seven (7) learning areas as recommended by the *Presidential Working Party on Educational Reforms* (PWPER, 2023). These shall comprise **four compulsory learning areas** namely: English, Kiswahili, Essential Mathematics/Core Mathematics and Community Service Learning (CSL). Learners will choose between Core Mathematics (for learners pursuing the Science Technology Engineering and Mathematics Pathway) and Essential Mathematics (for learners pursuing the non-STEM pathways). The learner will select three learning areas depending on their career choice, aptitude, interest and personality with guidance by the career teacher at Senior School.

LEARNING OUTCOMES FOR SENIOR SCHOOL

By the end of Senior School, the learner should be able to:

1. Communicate effectively and utilise information and communication technology across varied contexts.
2. Apply mathematical, logical and critical thinking skills for problem solving.
3. Apply basic research and scientific skills to manipulate the environment and solve problems.
4. Exploit individual talents for leisure, self-fulfilment, career growth, further education and training.
5. Uphold national, moral and religious values and apply them in day-to-day life.
6. Apply and promote health care strategies in day-to-day life.
7. Protect, preserve and improve the environment for sustainability.
8. Demonstrate active local and global citizenship for harmonious co-existence.
9. Demonstrate appreciation of diversity in people and cultures.
10. Manage pertinent and contemporary issues responsibly.

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LIST OF SUBJECTS AT SENIOR SCHOOL

Core Subjects	Arts & Sports Science	Social Sciences	Science, Technology, Engineering & Mathematics (STEM)
1. English 2. Kiswahili/KSL 3. Core Mathematics/ Essential Mathematics 4. Community Service Learning (CSL)	5. Sports and Recreation 6. Music and Dance 7. Theatre and Film 8. Fine Arts	9. Literature in English 10. Indigenous Languages 11. Fasihi ya Kiswahili 12. Sign Language 13. Arabic 14. French 15. German 16. Mandarin Chinese 17. Christian Religious Education 18. Islamic Religious Education 19. Hindu Religious Education 20. Business Studies 21. History and Citizenship 22. Geography	23. Biology 24. Chemistry 25. Physics 26. General Science 27. Agriculture 28. Computer Studies 29. Home Science 30. Aviation 31. Building Construction 32. Electricity 33. Metalwork 34. Power Mechanics 35. Wood Technology 36. Media Technology 37. Marine and Fisheries Technology

Note:

a) Physical Education (PE) and Information Communication and Technology (ICT) will be offered to all learners to facilitate learning, enjoyment and acquisition of life skills.

b) Every school shall offer Pastoral/Religious Programme of Instruction (P/RPI) to enhance moral, spiritual and character development of the learners.

c) There shall be One lesson for personal or group study to deepen knowledge and enhance skills in areas of choice by the learners.

Table 2 below provides guidelines on distribution of lessons at Senior School.

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LESSON DISTRIBUTION AT SENIOR SCHOOL

Subjects	No of Lessons Per Week (40 minutes per lesson)
Core Subjects 4	
1. English	5
2. Kiswahili	5
3. Essential Mathematics/Core Mathematics	5
4. Community Service Learning	3
Elective Subjects	
5. Elective 1	5
6. Elective 2	5
7. Elective 3	5
* *Physical Education*	*3*
* *ICT Skills*	*2*
**PPI*	*1*
* *Learner Personal /Group Study*	*1*
Total Number of Lessons	40

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ESSENCE STATEMENT

Power Mechanics as a subject build upon the competencies acquired in pre-technical studies at Junior School. At this level, learners will deepen their understanding of mechanical principles, power systems and practical applications moving from basic concepts to more complex and specialized ones.

The subject covers key content areas including; Fundamentals of power mechanics, Related drawing, Vehicle systems and Engines. The transition emphasises enhanced critical thinking, problem-solving, and hands-on skills that prepare learners for real-world technical challenges. Learners are expected to develop the core competencies and values outlined in the Basic Education Curriculum Framework (BECF). Pertinent and contemporary issues (PCIs) such as social economic and environmental issues and citizenship education and are mainstreamed in the learning experiences. Learning at this level is learner cantered, and formative assessment emphasised.

Upon completion of Senior School, competencies gained by learners in this subject will enable them transit into various post-secondary tertiary institutions such as Universities, National polytechnics, technical training institutes, Technical and vocational centres or transit to the world of work.

SUBJECT GENERAL LEARNING OUTCOMES

By the end of Senior School, the learner should be able to:

1. Attain a firm foundation for career growth, further training and education in power mechanics related areas of study
2. Utilise acquired power mechanics competencies to service and repair motor vehicle systems
3. Apply acquired drawing skills to interpret and communicate engineering designs using information and communication technology where applicable.
4. Relate positively with members of the society during visits to the work environment.
5. Use acquired knowledge to identify business opportunities related to power mechanics
6. Appreciate the place of power mechanics in day-to-day life

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SUMMARY OF STRANDS AND SUB STRANDS

Strand	Sub Strand
1.0. Fundamentals of Power Mechanics	1.1. Overview of Power Mechanics as a learning area 1.2. Evolution of motor vehicles 1.3. Power mechanics workshop layout 1.4. General workshop rules and regulations
2.0 Related Technical Drawing	2 1. Diagonal scale 2 2. Loci 2 3. Tangency 2 4. Blending of lines and curves
3.0 Motor Vehicle Systems	3.1 Road wheels 3.2 Motor Vehicle body 3.3 Motor Vehicle chassis 3.4 Motor Vehicle body Joining processes
4.0 Engines	4.1 Introduction to engines 4.2 Types of Engines 4.3 Classifications of Engine 4.4 Engine Components

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STRAND 1.0: FUNDAMENTALS OF POWER MECHANICS

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
1.0 Fundamentals of Power Mechanics	**1.1 Overview of Power Mechanics as a learning area**	By the end of the sub strand the learner should be able to: a) describe power mechanics as a learning area, b) identify career opportunities in power mechanics field, c) evaluate activities related to power mechanics in the locality, d) acknowledge the importance of power mechanics as a learning area.	The learner is guided to: ● use online or print resources to search for information on definition and scope of power mechanics as a learning area, ● brainstorm on career opportunities related to power mechanics, ● explore the local community to identify activities related to power mechanics, ● discuss on the importance of power mechanics as a learning area, ● engage a resource person on the importance of power mechanics as a learning area.	How is power mechanics important in day-to-day life?

Core competencies to be developed:

- Learning to learn: As the learner develops skills of organising own learning while searching for information from learning resources on the definition and scope of power mechanics as a learning area.
- Communication and Collaboration: As the learner develops skills of speaking clearly and effectively while discussing the importance of power mechanics as a learning area,

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- Digital literacy: As the learner develops skills of interacting with digital technology while accessing online resources to search for definition and scope of power mechanics as a learning area.
- Citizenship: As the learner develops a sense of belonging to own country while exploring the local community to identify activities related to power mechanics.

Pertinent and Contemporary issues (PCIs):

- Decision making: As the learner explores the local community to identify activities related to power mechanics.
- Career guidance: As the learner brainstorms on career opportunities related to power mechanics.
- Online safety: As the learner securely makes reference to online resources while searching for information on definition and scope of power mechanics as a learning area.

Values:

- Unity: As the learner cooperates with peers and displays team spirit while brainstorming.
- Respect: As the learner displays open mindedness and appreciates diverse opinions while discussing.
- Patriotism: As the learner develops a sense of belonging to own country while exploring the local community to identify activities related to power mechanics.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
1.0 Fundamentals of Power Mechanics	**1.2. Evolution of motor vehicles**	By the end of the sub strand, the learner should be able to: a) explain the evolution of motor vehicles in global perspective, b) describe trends in the historical developments of motor vehicles, c) explore the importance of innovation in motor vehicle development, d) appreciate the importance of the evolution of the motor vehicle.	The learner is guided to: ● brainstorm on evolution of motor vehicle globally, ● use digital or print resources to search for information on the history and trends of motor vehicle development, ● use online resources and print media to search for key trends in the historical developments of motor vehicles, ● discuss the importance of innovation in motor vehicle development, ● reflect on the importance of the evolution of motor vehicles.	How has technology advancement affected motor vehicles?
Core competencies to be developed: ● Communication and Collaboration: As the learner develops eloquence and togetherness while discussing with peers the importance of innovation in motor vehicle development. ● Self-efficacy: As the learner develops skills of articulating their views clearly during the discussion.				

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• Digital literacy: The learner develops skills of connecting using technology while accessing online resources to search for information on trends in motor vehicle development.
Pertinent and contemporary issues (PCIs): • Social awareness skills: As the learner develops assertiveness while engaging in discussion with peers on the importance of innovation in motor vehicle development. • Online safety: As the learner shows accountability in own actions while using online resources to search for information on key trends in the historical developments of motor vehicles. • Self-management skills: As the learner develops self-esteem while engaging with peers in a discussion on the importance of innovation in motor vehicle development.
Values: • Unity: As the learner cooperates and displays team spirit while brainstorming with peers on the evolution of motor vehicles globally. • Respect: As the learner develops a sense of acceptance when accepting diverse opinions during the discussion on the importance of innovation in motor vehicle development. • Social justice: As the learner shares digital and online resources equitably while searching for information on the history and trends of motor vehicle development.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
1.0 Fundamentals of Power Mechanics	**1.3. Power Mechanics workshop layout**	By the end of the sub strand, the learner should be able to: a) identify main areas in a power mechanics workshop, b) sketch a layout of main areas in a power mechanics workshop, c) explain the importance of various workshop areas in power mechanics, d) interpret the floor markings used in a power mechanics workshop, e) appreciate the importance of the workshop layout in power mechanics.	The learner is guided to: ● use online resources and print media to search for information on the main areas of power mechanics workshop (*storage, working, work benches, office, passage ways among others*), ● visit a power mechanics workshop in the locality to identify the main areas, ● brainstorm on the main areas of a power mechanic workshop, ● locate the main areas of power mechanics workshop and indicate their purposes, ● make illustrations of standard power mechanics workshop layout, ● use online resources and print media to search for information on the importance of various workshop areas in power mechanics,	Why is a power mechanics workshop layout important?

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			• use available resources to search for the information on the interpretation of floor markings used in a power mechanics workshop, • brainstorm on floor markings are used to enhance safety on roads (*pedestrian crossing, bus stage, lane lines, carriage boundaries etc)* • use models of roads, highways or other available resource identify markings on road surfaces, (*pedestrian crossing, bus stage, lane lines, carriage boundaries etc)* • engage a resource person in a discussion on interpretations of road surface markings, • acknowledge the importance of the workshop layout in power mechanics.	
Core competencies to be developed: • Critical thinking and Problem solving: As the learner develops evaluation and decision making skills while locating the main areas of a power mechanic workshop and indicates their purposes. • Communication and Collaboration: As the learner develops listening and speaking skills while brainstorming on the main areas of a power mechanic workshop.				

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- Self-efficacy: As the learner develops self-awareness skills while managing given tasks and shows concerted attention to detail when successfully makes illustrations of standard power mechanics workshop layout.
- Learning to learn: The learner searches for information on the main areas of the power mechanics workshop.
- Citizenship: As the learner develops a sense of belonging to his/her own country while visiting a power mechanics workshop in the locality to identify the main areas.

Pertinent and contemporary issues (PCIs):

- Assertiveness: As the learner actively participates and airs own views while brainstorming.
- Safety in the workshops: As the learner searches for the information on the interpretation of floor markings used in a power mechanics workshop.
- Online safety: As the learners display a sense of accountability while using online resources to search for information on the main areas of a power mechanic workshop.
- Road safety: As the learner uses models of roads, highways or other available resources to identify markings on surfaces.

Values:

- Peace: As the learner displays love and respect while brainstorming on the main areas of a power mechanic workshop.
- Respect: As the learner develops a sense of acceptance and appreciates diverse opinions from peers while brainstorming on other areas where floor markings are used to enhance safety.
- Patriotism: As the learner develops a sense of being a citizen of his/her county while visiting a power mechanics workshop in the locality to identify the main areas.

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Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
1.0 Fundamentals of Power Mechanics	**1.4. General workshop rules and regulations**	By the end of the sub strand the learner should be able to: a) describe general safety rules and regulations in power mechanics workshop, b) interpret workshop safety signs used in power mechanics environment, c) explain risk behaviours in a workshop environment, d) practise safe use of personal protective equipment (PPE) in power mechanics workshop, e) appreciate the need for safety while working in a power	The learner is guided to: ● brainstorm with peers the general safety rules and regulations in power mechanics work environment, ● use digital and print resources to search for information on safety rules and regulations in power mechanics workshop, ● use visual aids to interpret workshop safety signs used in power mechanics environment (*prohibition, mandatory, warning, and fire equipment signs)* ● brainstorm on risk behaviours in power mechanics workshop environment, ● discuss with pears on the risk behaviours in the environment (*playing on the road, drivers speeding, jay walking across the road etc*)	Why is safety important in a power mechanic work environment?

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		mechanics work environment.	• use online resources and print media to search for uses of various personal protective equipment (PPE) as per work environment rules and regulations, • use online and print resources to search for information on other areas in the environment where use of PPEs is required (*while using roads-reflective strips, clothing)* • use personal protective equipment (PPE) according to work environment rules and regulations, • acknowledge the need for safety while working in a power mechanics work environment.	
Core competencies to be developed: • Critical thinking and Problem solving: As the learner develops evaluation skills while using visual aids to interpret workshop safety signs. • Self-efficacy: As the learner confidently presents own views while discussing the general safety rules and regulations in a power mechanics work environment. • Digital literacy: As the learner develops skills of manipulating digital technology while using online resources to search for information on personal protective equipment (PPE) in a power mechanics work environment.				

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● Learning to learn: As the learner searches for information on safety rules and regulations in a power mechanics workshop.
Pertinent and contemporary issues (PCIs): ● Assertiveness: As the learner airs their views with confidence while discussing general safety rules and regulations in a power mechanics work environment. ● Safety and security: As the learner brainstorms with peers the general work environment rules and regulations. ● Online safety: As the learner accounts for own actions while using online resources to search for uses of various personal protective equipment (PPEs).
Values: ● Unity: As the learner cooperates and collaborates while discussing the risk behaviours in the environment. ● Love: As the learner puts the interest of others before their own while they share resources to search for information on safety rules and regulations in power mechanics workshop, ● Responsibility: As the learner develops a sense of self-drive on observing safety while using personal protective equipment (PPE) according to work environment rules and regulations. ● Respect: As the learner appreciates diverse opinions and develops a sense of acceptance while discussing.

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Suggested Assessment Rubric				
Level / **Indicator**	**Exceeds Expectations**	**Meets Expectation**	**Approaches Expectations**	**Below Expectations**
Ability to describe power mechanics as a learning area.	Comprehensively describes power mechanics as a learning area.	Describes power mechanics as a learning area.	Describes power mechanics as a learning area leaving out few key words.	Describes power mechanics as a learning area leaving out many keywords.
Ability to describe trends in the historical developments of motor vehicles.	Comprehensively describes trends in the historical developments of motor vehicles.	Describes trends in the historical developments of motor vehicles.	Describes trends in the historical developments of motor vehicles skipping a few.	Describes trends in the historical developments of motor vehicles skipping many.
Ability to sketch a layout of main areas in a power mechanics workshop.	Creatively sketches a layout of the main areas in a power mechanics workshop.	Sketches a layout of the main areas in a power mechanics workshop.	Sketches a layout of the main areas in a power mechanics workshop leaving out a few.	Sketches a layout of the main areas in a power mechanics workshop leaving out many.
Ability to practise safe use of personal protective equipment (PPE) in a power mechanic workshop.	Consistently practises safe use of personal protective equipment (PPE) in a power mechanic workshop.	Practises safe use of personal protective equipment (PPE) in a power mechanic workshop.	Practises safe use of most of the personal protective equipment (PPE) in a power mechanic workshop.	Practises use of only a few of the personal protective equipment (PPE) in a power mechanic workshop.

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STRAND 2.0: RELATED DRAWING

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
2.0 Related Drawing	**2 1.Scales**	By the end of the sub strand the learner should be able to: a) identify technical drawing instruments, b) describe diagonal scale used in drawing, c) construct diagonal scale used in drawing, d) interpret diagonal scale used in drawing, e) appreciate the application of diagonal scale in real life.	The learner is guided to: ● use available resources to search for information on technical drawing instruments ((*set squares-30º.45º.60º and 90º degrees; compass; divider, drawing board, and T-square*), ● use online resources and print media to search for information on diagonal scale, ● practise drawing diagonal scale, ● brainstorm on the interpretation of diagonal scale used in drawing, ● acknowledge the application of diagonal scale in real life.	Why are scales in drawing important in real life?
Core competencies to be developed: ● Critical thinking and Problem solving: While the learner practises drawing diagonal scale. ● Learning to learn: As the learner develops researching skills while searching for information on diagonal scale. ● Digital literacy: As the learner manipulates digital devices to access online resources while searching for information on diagonal scales.				

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Pertinent and contemporary issues (PCIs): • Effective communication: As the learner clearly airs own views while brainstorming on the interpretation of diagonal scale used in drawing. • Online safety: As the learner responsibly makes reference to the internet while searching for information on diagonal scale. • Assertiveness: While leaner decisively draws diagonal scale.
Values: • Unity: As the learner cooperates and strives to achieve common goals while brainstorming on the interpretation of diagonal scale used in drawing. • Respect: As the learner displays open mindedness and appreciates diverse opinions while brainstorming.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
2.0 Related Drawing	**2 2. Loci**	By the end of the sub strand, the learner should be able to: a) define a loci as used in drawing, b) describe types of loci used in drawing, c) construct different types of loci used in technical drawing, d) construct the locus of link mechanisms relevant to power mechanics, e) appreciate the application of loci in power mechanics.	The learner is guided to: ● use online and print resources to search for the definition of loci of a point in drawing, ● use online resources and print media to search for the description of types of loci, ● practise drawing types of loci (*ellipse, cycloid, involute, parabola, hyperbola),* ● discuss the procedure of drawing the locus of a mechanism, ● practise drawing the loci of a point on a mechanism (*3-link and 2-link),* ● discuss the application of loci in power mechanics.	1. How are loci used in real life situations? 2. How are skills on loci applied in power mechanics?
Core competencies to be developed: ● Critical thinking and Problem solving: As the learner develops skills of navigating complex situations while drawing the locus of a point on a given mechanism. ● Communication and Collaboration: As the learner develops teamwork skills while making a contribution and listens during the discussion. ● Self-efficacy: As the learner develops a sense of belief in self while constructing different loci successfully.				

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• Digital literacy: As the learner interacts with digital devices to access online resources while searching for information on types of loci.
Pertinent and contemporary issues (PCIs): • Effective communication: As the learner develops listening and speaking skill while discussing the applications of loci in power mechanics. • Online safety: As the learner displays a sense of accountability while using online resources to search for the description of types of loci. • Empathy: As the learner connects with each other while engaging in a discussion on the procedure of drawing the locus of a mechanism.
Values: • Social justice: As the learner equitably shares digital and print resources used while searching for information. • Unity: As the learner cooperates and displays team spirit while discussing the applications of loci in power mechanics. • Respect: As the learner appreciates diverse opinions while discussing the application of loci in power mechanics.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
2.0 Related Drawing	**2 3.Tangency**	By the end of the sub strand, the learner should be able to: a) describe the methods of constructing tangents in drawing, b) construct a tangent to a circle from a point, c) construct external tangent to two circles, d) construct interior tangent to two circles, e) appreciate the importance of tangency in drawing.	The learner is guided to: ● use online or print resources to search for information on tangency as used in drawing, ● discuss with peers the methods of constructing tangents, ● practise drawing a tangent to a circle from a point outside the circle, ● practise drawing external tangent to two circles *(equal circles, unequal circles),* ● practise drawing an internal tangent to two circles *(equal circles, unequal circles)*, ● discuss the importance of tangency in drawing.	Where is tangency applied in real life situations?

Core competencies to be developed:

- Critical thinking and Problem solving: As the learner develops analytical skills while drawing interior and external tangents.
- Communication and Collaboration: As the learner develops teamwork skills while participating in a conversation and strives for a common goal during the discussion.
- Self-efficacy: As the learner develops a sense of self belief while drawing tangents successfully.
- Digital literacy: As the learner develops skills of interacting with digital devices while accessing online resources to search for information on tangents.

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Pertinent and contemporary issues (PCIs):

- Effective communication: As the learner makes own contribution and peers listen while discussing the methods of constructing tangents.
- Assertiveness: As the learner successfully draws a tangent to a circle from a point outside the circle.
- Self-awareness: As the learner manages emotional intelligence while discussing with peers on the methods of constructing tangents.

Values:

- Social justice: As the learner shares resources equitably while practicing drawing external tangent to two circles.
- Unity: As the learner cooperates and strives to achieve common goals while discussing with peers.
- Respect: While the learner accepts diverse opinions while discussing.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
2.0 Related Drawing	**2.4 Blending of lines and curves**	By the end of the sub strand the learner should be able to: a) describe blending of lines and arcs as used in drawing, b) construct arcs that blend lines, c) construct arcs that blend two circles, d) use blending to construct given shapes, e) appreciate the application of blending in power mechanics.	The learner is guided to: • use online or print resources to search for information on blending of lines and arcs in drawing, • discuss on the procedure of blending curves and lines, • blend a curve and a line (*two straight lines, straight line and circle),* • blend two equal circles (internal *and external*), • construct different shapes using blending, • brainstorm on the applications of blending of lines and curves in power mechanics.	1. Why is blending of lines and curves important in drawing? 2. How is blending of lines and curves applied in real life?

Core competencies to be developed:

- Critical thinking and Problem solving: As the learner develops high order thinking skills while constructing different shapes using blending.
- Self-efficacy: As the learner successfully blends two equal circles.
- Digital literacy: As the learner develops skills of connecting with technology while interacting with digital devices to access online resources to search for information.

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Pertinent and contemporary issues (PCIs):

- Assertiveness: As the learner selects the correct method of blending arcs and lines.
- Effective communication: As the learner makes own contribution and peers listen while brainstorming on the applications of blending of lines and curves in power mechanics.
- Online safety: As the learner responsibly accesses online resources to search for information on blending of lines and arcs in drawing.

Values:

- Unity: As the learner cooperates and displays team spirit while brainstorming on the applications of blending of lines and curves in power mechanics.
- Respect: As the learner shows open mindedness and appreciates diverse opinions while discussing on the procedure of blending curves and lines.

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Suggested Assessment rubric				
Level / Indicator	**Exceeds Expectations**	**Meets Expectations**	**Approaches Expectations**	**Below Expectations**
Ability to describe diagonal scale used in drawing.	Comprehensively describes diagonal scale used in drawing.	Describes diagonal scale used in drawing.	Describes most of the diagonal scale used in drawing.	Describes only a few of the diagonal scale used in drawing.
Ability to construct different types of loci used in technical drawing.	Accurately constructs different types of loci used in technical drawing.	Constructs different types of loci used in technical drawing.	Constructs different types of loci used in technical drawing leaving few details.	Constructs different types of loci used in technical drawing leaving many details.
Ability to construct interior tangent to two circles.	Accurately constructs interior tangent to two circles.	Constructs interior tangent to two circles.	Partially constructs interior tangent to two circles.	Constructs interior tangent to two circles with assistance.
Ability to construct arc (s) that blend two circles.	Neatly constructs arc (s) that blend two circles.	Constructs arc (s) that blend two circles.	Partially constructs arc (s) that blend two circles.	Constructs arc (s) that blend two circles with assistance.

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STRAND 3.0: MOTOR VEHICLE SYSTEMS

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
3.0 Motor Vehicle Systems	**3.1 Road wheels**	By the end of the sub strand the learner should be able to: a) explain the functions of wheels in a vehicle, b) identify types of tires used in vehicles, c) illustrate types of ply orientation used on vehicle tires, d) interpret specifications on tire sidewalls in relation to road safety, e) appreciate the role of tires in vehicle operations.	The learner is guided to: ● use digital and print resources to search for information on functions of wheels in motor vehicles, ● brainstorm on the types of vehicle tires, (*tube and tubeless)* ● sketch types of tire ply orientations, (*radial, cross ply)* ● use digital or print resources to search for information on tires specification and their interpretation, (*size, width, tire type, aspect ratio, load capacity, speed)* ● interpret tire specification on a given tire wall and relate to road safety, ● identify various types of abnormal tire wear, ● use online and print resources to search for information on the	Why is it important to study about road wheels?

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			safety hazards associated to failure to adhere to tire specifications, • perform basic tire maintenance (puncture repair, tire pressure checks) • value the role of tires in vehicles.	

Core competencies to be developed:

- Learning to learn: As the learner develops researching skills while making reference to digital and print resources to search for information on motor vehicle road wheels.
- Communication and Collaboration: As the learner develops teamwork skills while making own contribution and peers listen when brainstorming on the types of vehicle tires.
- Digital literacy: As the learner interacts with digital devices while searching for information on functions of tires.

Pertinent and contemporary issues (PCIs):

- Effective communication: As the learner makes a contribution and peers listen while brainstorming on types of vehicle tires.
- Decision making: As the learner interprets tire specifications.

Values:

- Unity: As the learner displays team spirit while brainstorming on types of tires.
- Respect: As the learner accommodates diverse opinions while brainstorming types of vehicle tires.
- Love: As the learner displays selflessness and puts the interest of others before own while sharing learning resources to search for information on functions of wheels in vehicles.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
3.0 Motor Vehicle Systems	**3.2 Vehicle body**	By the end of the sub strand the learner should be able to: a) describe functions of a body in a motor vehicle, b) illustrate types of bodies used in motor vehicles, c) classify the various types of vehicles according to their uses, d) identify interior body parts of a vehicle, e) illustrate exterior body parts of a motor vehicle, f) appreciate the importance of various body parts in a motor vehicle.	The learner is guided to: ● brainstorm on the functions of vehicle body, ● use digital and print resources to search for information on vehicle body types (*salon, station wagon, convertible, van, pickup and bus and others),* ● practise sketching types of bodies used in motor vehicles, ● develop a chart on various types of vehicles according to their uses (*emergency response vehicles, passenger service vehicles-psv, commercial vehicles*) ● discuss the types of emergency vehicles in their locality and note their emergency response numbers, ● identify roles of emergency response vehicles (*ambulances, police vehicles, fire engines*) ● visit the locality to explore on different interior parts of a vehicle *(seats, floor, roof, doors, door handles, seats, dashboard, safety belts),*	Why is it important to learn about vehicle body and body parts?

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			• use visual aids to search for information on exterior vehicle body parts, *(bumpers, fenders, roofs, bonnet, sills, door, boot, side wings, roof, roof rack, roof rails).* • engage a resource person to explain the interior and exterior body parts of a vehicle that are associated with safety (*seat belt, air bags, bumpers, head rest, arm rest, dashboard padding, child safety seats, cargo barriers etc.)* • acknowledge the importance of various body parts in a motor vehicle.	

Core competencies to be developed:

- Learning to learn: As the learner develops skills of organising own learning while searching for information on functions of a vehicle body.
- Communication and Collaboration: As the learner makes contributions and listens to peers while discussing the types of emergency vehicles in their locality.
- Digital literacy: As the learner develops skills of manipulating digital devices to access online resources while searching for vehicle body types.

Pertinent and contemporary issues (PCIs):

- Online safety: As the learner shows accountability for own actions while accessing the internet to search for information on vehicle body types.
- Assertiveness: As the learner successfully sketches the types of bodies used in motor vehicles.
- Self-awareness: As the learner develops a chart on various types of vehicles according to their uses.

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Values:

- Unity: As the learner cooperates and displays a team spirit while brainstorming on the functions of a vehicle body.
- Love: As the learner selflessly puts interest of others before own while sharing learning resources to search for information on vehicle body types.
- Respect: As the learner appreciates diverse opinions while discussing.
- Responsibility: As the learner displays self-drive and engages in assigned roles while practising sketching of types of bodies used in motor vehicles.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
3.0 Motor Vehicle Systems	**3.3 Motor Vehicle Chassis**	By the end of the sub strand the learner should be able to: a) describe types of chassis used in motor vehicles, b) explain the functions of chassis in a vehicle, c) illustrate different types of frame sections used in motor vehicle chassis, d) identify safety features in vehicle chassis, e) appreciate the role of motor vehicle chassis.	The learner is guided to: ● use print and online resources to search for types of chassis used in motor vehicles (*separate, integral)* ● visit the locality to identify types of motor vehicle chassis, ● discuss with peers the functions of chassis in a motor vehicle, ● sketch different types of frames used in construction of motor vehicle chassis *(tube, channel, box, flat).* ● use visual aids to identify safety features in motor vehicle chassis used to minimise damage and injuries during a crash, (*collapsible points and dampers)* ● acknowledge the role of motor vehicle chassis.	Why is chassis important in a motor vehicle body?
Core competencies to be developed: • Learning to learn: As the learner develops researching skills while searching for information on the types of chassis used in motor vehicles. • Self-efficacy: As the learner correctly sketches different types of frames used in construction of motor vehicle chassis.				

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● Digital literacy: As the learner interacts with digital devices while accessing online resources to search for information.
Pertinent and contemporary issues (PCIs): ● Effective communication: As the learner clearly makes own contribution and peers listen during the discussion on the functions of chassis in a motor vehicle. ● Social cohesion: As the learner connects to the society as they visit to identifying different types of motor vehicle chassis.
Values: ● Unity: As the learner cooperates and displays team spirit while discussing with peers on the function of motor vehicle chassis. ● Respect: As the learner appreciates diverse opinions while discussing the functions of motor vehicle chassis. ● Patriotism: As the learner develops a sense of belonging to own country while visiting the locality to identify types of motor vehicle chassis.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
3.0 Motor Vehicle Systems	**3.4 Vehicle body joining processes**	By the end of the sub strand the learner should be able to: a) describe threaded fasteners used in motor vehicle body joining, b) illustrate locking devices uses in motor vehicle body, c) perform riveting on a motor vehicle body, d) perform joining process on motor vehicle body parts using adhesive, e) acknowledge the role of joining processes in a motor vehicle body.	The learner is guided to: ● visit the locality to identify joining processes used in motor vehicles, ● brainstorm on the joining processes used in motor vehicles, ● use online resources and print media to search for information on threaded fasteners used in motor vehicle body joining (*bolts and nuts, screws, studs*), ● identify and sketch the major categories of threaded fasteners (*screws; phillips head, flathead, hex and torx. Bolds: hex, carriage. Nuts; hex, flange, slotted, wing, dome*), ● Sketch a threaded fastener and label (*diameter, pitch, head, length, thread angle, root and crest),* ● use online and print media resources to search for the	How are joining processes applied in motor vehicle bodies?

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			information on locking devices (*cotter pin, circlip and split pin*), • identify locking devices used in motor vehicle body in the locality and sketch them, • use online and print media to search information on types of rivets, riveting process, riveting tools, application of riveting in motor vehicle body, • carry out given joining process on motor vehicle body parts using adhesive, • use online resources to search for information on definition of the terms used in adhesives (*adhesive, bond, cohesion*). • appreciate the role of joining processes in a motor vehicle body.	

Core competencies to be developed:

- Digital literacy: As the learner develops skills of connecting with digital technology while using online resources to search for information on riveting process, riveting tools, and application of riveting in a motor vehicle body.
- Self-efficacy: As the learner successfully carries out a given joining process on motor vehicle body parts using adhesive.
- Learning to learn: As the learner develops skills of organising own learning while searching for information on threaded fasteners used in motor vehicle body joining.

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Pertinent and contemporary issues (PCIs): • Self-esteem: As the learner carries out a given joining process on motor vehicle body parts using adhesive. • Effective communication: As the learner makes own contribution while brainstorming on the joining processes used in motor vehicles.
Values: • Responsibility: As the learner engages in assigned roles and duties while carrying out a given joining process on motor vehicle body parts using adhesive. • Respect: As the learner appreciates diverse opinions while brainstorming. • Patriotism: As the learner identifies self with own country and develops love for it while visiting the locality to identify joining processes used in motor vehicles.

Suggested Assessment rubric

Level / Indicator	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to interpret specifications on tire sidewalls in relation to road safety.	Exhaustively interprets specifications on tire sidewalls in relation to road safety.	Interprets specifications on tire sidewalls in relation to road safety.	Interprets specifications on tire sidewalls in relation to road safety leaving a few.	Interprets specifications on tire sidewalls in relation to road safety leaving many.
Ability to illustrate different types of bodies used in motor vehicles.	Creatively illustrates different types of bodies used in motor vehicles.	Illustrates different types of bodies used in motor vehicles.	Illustrates different types of bodies used in motor vehicles leaving few details.	Illustrates different types of bodies used in motor vehicles leaving many details.

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Level / Indicator	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to illustrate different types of frame sections used in motor vehicle chassis.	Illustrates types of frame sections used in motor vehicle chassis in detail.	Illustrates types of frame sections used in motor vehicle chassis.	Illustrates types of frame sections used in motor vehicle chassis leaving few details.	Illustrates types of frame sections used in motor vehicle chassis leaving many details.
Ability to perform riveting on a motor vehicle body.	Performs riveting on a motor vehicle body without defects.	Performs riveting on a motor vehicle body with one defect.	Performs riveting on a motor vehicle body with two defects.	Performs riveting on a motor vehicle body with three defects.

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STRAND 4.O: ENGINES

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
4.0 Engines	**4.1 Introduction to Engines**	By the end of the sub strand the learner should be able to: a) define an engine as used in power mechanics, b) identify uses of engines in day-to-day life, c) explain terminologies used in engines, d) calculate the capacity of a given engine, e) appreciate the importance of engines in day to-day-life.	The learner is guided to: ● use online or print resources to search for definitions of an engine as used in power mechanics, ● discuss with peers the definitions of an engine as used in power mechanics, ● use digital or print media to search for the uses of engines in day-to-day life, ● sketch a sectioned single cylinder engine and name the major the major parts (*block, piston, cylinder, valves and cylinder head*), ● brainstorm on the terminologies used in engines, ● use online or print resources to search for information on engine terminologies (*bore, stroke, capacity, horse power, TDC, BDC*),	How are engines used in day-to-day life?

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			• carry out measurement and calculation to determine capacity of an engine using bore and stroke, • acknowledge the importance of engines in day to-day- life.	

Core Competencies to be developed:

- Communication and Collaboration: While the learner develops teamwork skills while brainstorming.
- Self-efficacy: As the learner successfully carries out measurements and calculates the capacity of an engine.
- Digital literacy: As the learner manipulates digital devices to access online resources while searching for definitions of engine terminologies.

Pertinent and contemporary issues (PCIs):

- Effective communication skills: As the learner brainstorms on the meaning of the terminologies used in engines.
- Online safety: As the learner accesses online resources to search for definitions of engine terminologies.
- Assertiveness: As the learner decisively carries out various measurements on an engine.

Values:

- Unity: As the learner cooperates and strives to achieve common goals while discussing the definitions of an engine as used in power mechanics.
- Respect: As the learner accommodates others' opinions while brainstorming on the terminologies used in engines.
- Social justice: As the learner shares resources while carrying out measurement and calculation to determine capacity of an engine using bore and stroke.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
4.0 Engines	**4.2 Classifications of Engines**	By the end of the sub strand the learner should be able to; a) explain methods of classifying engines in power mechanics, b) categorize given engines into different classes, c) value the different classes of engines.	The learner is guided to: ● use digital or print media to search for information on the methods of classifying engines (*method of combustion, size; small, medium, large*, *type of fuel; petrol, diesel and alternative fuel, cooling; air and liquid)* ● visit power mechanics related workplaces in the locality to identify various classes of engines, ● engage in a discussion with peers on engine classification, ● generate a table and put given engines into respective classes, ● acknowledge the different classification of engines.	How are engines classified?
Core competencies to be developed: ● Learning to learn: As the learner develops researching skills while searching for information online. ● Self-efficacy: While the learner successfully generates a table and puts given engines into respective classes.				

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• Citizenship: As the learner develops a sense of belonging to own country while visiting power mechanics related workplaces in the locality to identify various classes of engines. • Digital literacy: As the learner uses digital resources to search for information on the methods of classifying engines.
Pertinent and contemporary issues (PCIs): • Assertiveness: As the learner generates a table and puts given engines into respective classes. • Discipline: As the learner respectfully works with peers during discussions. • Online safety: As the learner responsibly accesses the internet while searching for information on the methods of classifying engines.
Values: • Patriotism: As the learner develops a sense of citizenship while visiting power mechanics related workplaces in the locality to identify various classes of engines. • Unity: As the learner cooperates and displays team spirit while discussing. • Respect: As the learner appreciates diverse opinions while discussing.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
4.0 Engines	**4.3 Types of Engines**	By the end of the sub strand the learner should be able to; a) explain the types of engines based on combustion process, b) describe the type of engines based on number of cylinders, *c)* illustrate the types of engines based on cylinder configuration, *d)* explain the types of engines based on operation characteristics, e) illustrate types of engines based on number of strokes, f) appreciate the types of engines used in day-to-day life.	The learner is guided to: ● use online and print resources to search for information on types of engines based on the combustion process (*external combustion and internal combustion),* ● brainstorm on the types of engines based on the number of cylinders (*single cylinder, multi cylinder),* ● use online and print resources to search for information about the types of engines based on cylinder configuration (*inline, V, rotary),* ● sketch the types of engines based on cylinder orientation (*inline, V, rotary),* ● discuss with peers on the types of engines based on operation characteristics (*high speed, medium speed, low speed*), ● use illustrations to explain four-stroke and two-stroke engine cycles,	Why are engines designed and built differently?

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			• acknowledge the importance of the different types of engines.	

Core competencies to be developed:

- Critical thinking and Problem solving: As the learner develops analytical skills while using illustrations to explain four-stroke and two-stroke engine cycles.
- Learning to learn: When the learner develops skills of connecting with technology while using digital resources to search for information.
- Communication and Collaboration: As the learner listening and speaking skills while discussing with peers.
- Self-efficacy: While the learner successfully sketches the types of engines based on cylinder orientation.

Pertinent and contemporary issues (PCIs):

- Assertiveness: As the learner acquires skills of defending their position during discussion.
- Discipline: As the learner respectfully works with peers during discussions.
- Online safety: As the learner responsibly accesses the internet to search for information about the types of engines.

Values:

- Unity: As the learner cooperates and displays teamwork while brainstorming.
- Respect: As the learner appreciates diverse opinions during discussions.
- Love: As the learner puts the interest of others before their own interest when sharing online and print resources while searching for information.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Questions
4.0 Engines	**4.4 Engine components**	By the end of the sub strand, the learner should be able to: a) identify components of an engine, b) sketch components of an engine, c) describe functions of components of an engine, d) perform measurements on engine components, e) appreciate the need for engine components.	The learner is guided to: ● use online and print resources to search for information on components of an engine, ● use regalia, visual aids or print resources to identify engine components (*crankshaft, camshaft, piston, connecting rod, valves, cylinder block, sump, rocker arms, cylinder head, rocker cover*), ● illustrate components of an engine (*crankshaft, camshaft, piston, connecting rod, valves, cylinder block*), ● brainstorm on the functions of engine components, ● discuss with peers the function of engine components, ● use appropriate tools to carry out measurements on engine components (*journal, cylinder bore and clearances, piston ring gap*). ● acknowledge the need for various engine components.	1. What are the main components of an internal combustion engine? 2. What are the key functions of internal combustion engine parts?

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Core competencies to be developed: ● Learning to learn: When the learner develops researching skills while searching for information on components of an engine. ● Communication and Collaboration: As the learner develops teamwork skills while contributing to group discussion on the function of engine components. ● Self-efficacy: As the learner successfully uses appropriate instruments to carry out measurements on engine components.
Pertinent and contemporary issues (PCIs): ● Assertiveness: As the learner acquires skills of defending their position during brainstorming. ● Discipline: As the learner respectfully works with peers during discussions. ● Online safety: As the learner responsibly uses the internet to search for information on components of an engine.
Values: ● Unity: While the learner displays team spirit when brainstorming. ● Respect: As the learner appreciates diverse opinions during discussions. ● Peace: As the learner displays calmness while discussing with peers the function of engine components.

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Suggested Assessment rubric				
Level / **Indicator**	**Exceeds Expectations**	**Meets Expectations**	**Approaches Expectations**	**Below Expectations**
Ability to calculate the capacity of an engine.	Calculates the capacity of an engine following all the steps.	Calculates the capacity of an engine.	Calculates the capacity of an engine skipping few steps.	Calculates the capacity of an engine skipping many steps.
Ability to explain methods of classifying engines in power mechanics.	Explains all methods of classifying engines in power mechanics.	Explains most of the methods of classifying engines in power mechanics.	Explains only a few of the methods of classifying engines in power mechanics.	Explains only a few methods of classifying engines in power mechanics with assistance.
Ability to categorize given engines into different classes.	Categorises given engines into different classes and gives examples.	Categorises given engines into different classes.	Categorises given engines into different classes leaving out a few.	Categorises given engines into different classes leaving out many.
Ability to illustrate the types of engines based on cylinder configuration.	Illustrates all types of engines based on cylinder configuration.	Illustrates most of the types of engines based on cylinder configuration.	Illustrates a few types of engines based on cylinder configuration.	Illustrates only a few types of engines based on cylinder configuration with assistance.

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APPENDIX: LIST OF ASSESSMENT METHODS, LEARNING RESOURCES AND NON-FORMAL ACTIVITIES

Strand	Sub Strand	Suggested Assessment Methods	Suggested Learning Resources	Non-Formal Activities
1.0 Fundamentals of power mechanics technology	**1.1 Overview of Power Mechanics as a learning area**	• Written tests • Question and answer • Checklist	Manilla paper and marker pens to write definition of power mechanics, video clips on activities related to power mechanics.	Learners visit workplace in the locality to observe activities related to power mechanics.
	1.2 Evolution of Motor Vehicles	• Question and answer • Written tests • Checklist	Pictures on the evolution of motor vehicle, visual aids and print media on key trends in the historical developments of the automobile.	Explore the locality to observe trends in motor vehicle development.
	1.3 **Power mechanics workshop layout**	• Written test • Question and answer • Checklist • Observation	Pictures and plans on power mechanics workshop layout Drawing papers Pencils and rulers to draw simple workshop layouts.	Visit the local workplace/workshops to conduct sensitization on the need for good workshop layout.

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Strand	Sub Strand	Suggested Assessment Methods	Suggested Learning Resources	Non-Formal Activities
	1.4 **General workshop rules and regulations**	• Question and answer • Written tests • Observe learners as they write down general safety regulations • Checklist	Manilla paper and marker pens to write the general workshop rules and regulations.	Visit a garage within the locality to develop flash cards on general workshop safety regulations.
2.0 Related Drawing	2 1.**Diagonal scale**	• Written test • Observe learners as they develop diagonal scales • Question and answer	Visual aids on diagonal scales Drawing paper and drawing set to draw diagonal scales.	Learners to explore the community and identify applications of diagonal scales in real life.
	2 2.**Loci**	• Question and answer • Written tests • Rubric • Observe learners as they develop various types of loci	Drawing paper and drawing equipment to draw ellipse, cycloid, involute, parabola and hyperbola locus.	Learners to engage workers in various workplaces within the community on uses of loci.
	2 3.Tangency	• Question and answer • Checklist • Observe learners as they construct various types of tangents	Drawing paper and drawing set for drawing tangents Charts with	Learners to engage a resource person on the application of tangency in power mechanics.

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Strand	Sub Strand	Suggested Assessment Methods	Suggested Learning Resources	Non-Formal Activities
			illustrations on types of tangents.	
	2 4.Blending of lines and curves	• Written test • Question and answer • Observe learners as they blend lines and circles	Drawing paper and drawing set to draw blended lines and curves, Charts with illustrations on blended lines and curves	Learners to identify and list items at home or in the community that were designed using knowledge of blending of lines and curves.
3.0 Motor Vehicle Systems	**3.1 Road wheels**	• Question and answer • Written tests • Rubric • Checklist	Pictures on road wheels, Used vehicle tires with specifications on the tire walls,	Visit parking areas in the locality and identify the various types of tires used in vehicles.
	3.2 Motor Vehicle body	• Written test • Question and answer • Checklist	Vehicles in garages and parking areas,	Visit parking areas or garages in the locality and identify the various types of vehicle body designs.
	3.3 Motor Vehicle chassis	• Question and answer • Written test • Checklist	Pictures of various types of vehicle chassis.	Visit vehicle garages in the locality to

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Strand	Sub Strand	Suggested Assessment Methods	Suggested Learning Resources	Non-Formal Activities
				identify vehicle chassis types.
	3.4 Motor Vehicle body Joining processes	• Question and answer • Written tests • Practical work • Checklist • Rubric	Vehicle garages where minor vehicle joining processes are performed Resource person to talk about the use of body hand tools in performing joining processes on vehicle body.	Participate in performing basic vehicle body joining processes in workplaces within the locality.
4.0 **Engines**	4.1 Introduction to engines	• Question and answer • Written test • Checklist	Pictures or video clips on general engine terminologies Basic measuring tools for measuring engine components Stroke and bore.	Develop charts on engine definition and engine terminologies.
	4.2 Types of engines	• Question and answer • Written tests • Rubric • Checklist	Pictures, sketched illustration on engine construction and orientation Used sectioned engines .	Visit workplaces in the community to identify types of engines used in day-to-day life.

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Strand	Sub Strand	Suggested Assessment Methods	Suggested Learning Resources	Non-Formal Activities
	4.3 Classification of engines	• Written test • Rubric • Question and answer • Observation schedules	Pictures or videos on different classification of engines, engine technology text books Used engines sectioned and with most of the components.	Visit workplaces in the community to observe types of engines classified using various parameters.
	4.4 Engine Components	• Written test • Checklist • Question and answer	Dismantled engines, sectioned engines, pictures, illustrations.	Visit a power mechanics workplace in the locality to observe engine parts.

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